L8 Qualitive methods

Social Science

What it is?

Social science is the study of **human behaviour**, how people think, feel, act, and interact.

It looks at **societies**, **groups**, and **social processes**.

It often uses **interviews, observations, and qualitative data** to understand why people do what they do.

Example(s)

- Studying why people buy certain products
- Observing how students behave in a classroom
- Interviewing people about their experiences
- Research on cultures, politics, or families

Natural Science

What it is?

Natural science is the study of the **physical world**, such as nature, matter, energy, and living organisms.

It uses **experiments, measurements, and quantitative data** to find general laws and patterns in nature.

Example(s)

- Studying how electricity works
- Measuring the speed of falling objects
- Observing how plants grow under different conditions
- Research in physics, chemistry, or biology

Simulation theory of interpretation

What it is?

Simulation theory says that we understand other people by using our own mind as a model.

We imagine what we would think or feel in their situation, and this helps us guess their mental states.

Example(s)

- You see someone looking sad and imagine how you would feel in that moment, so you conclude they are sad.
- You watch someone struggle with a task and mentally "try it yourself" to understand what they might be thinking.

Theory theory of interpretation

What it is?

Theory theory says that we understand other people by forming a kind of mental theory about how minds work.

We collect evidence about their behaviour, make hypotheses about their beliefs or desires, and update these hypotheses when we get new

information.

Example(s)

- You see someone avoid a dog and infer they must believe dogs are dangerous.
- You watch how a friend reacts in many situations and build a general idea of what they usually want or believe, like a small scientific theory about their mind.

Belief desire explanation

What it is?

A belief–desire explanation says that an agent chooses **action A** instead of **action B** because she **desires outcome X** (and not **Y**) and **believes** that **A will lead to X**, while **B will lead to Y**.

Example(s)

- The agent chooses **A** because she wants **X**, and she believes **A → X** and **B → Y**.
- If the agent desires **X** and believes that **only A** brings about **X**, she will pick **A** over **B**.

What are mental states?

Introspective psychology

What it says?

Mental states are **inner experiences** such as thoughts, feelings, desires, and emotions.

We can know them by **looking inside our mind** through introspection.

Example(s)

- "I feel sad."
- "I want to win."
- "I believe it will rain."

Methodological Behaviorism

What it says?

Mental states should **not** be studied because they cannot be observed directly.

Science should only study **behaviour** (what people do), not what they think or feel.

Example(s)

- Instead of saying "She is angry," a behaviourist says:

  "She is shouting and walking fast," because these are observable.

Cognitive science

What it says?

Mental states **do exist**, like beliefs, desires, plans, and intentions.

But we should study them scientifically by connecting them to **observable behaviour** and using evidence not introspection.

Example(s)

- Inferring someone's belief from their choices.
- Using reaction times or errors in tasks to understand mental processes.

The intentional stance

What it is?

The intentional stance is a way of understanding and predicting an agent's behaviour by **treating the agent as if it has beliefs and desires**.

This method can be used for **humans, animals, or even machines**.

We assume the agent is **rational**, meaning it will try to achieve its goals based on what it believes.

Example(s)

- Predicting that a person will take an umbrella because they **believe** it will rain and **desire** to stay dry.
- Saying a dog goes to the kitchen because it **believes** food is there and **wants** to eat.
- Saying a chess computer "wants to protect its queen" because this helps us predict its next move, even though the computer does not literally have desires.

Qualitative data

What it is?

Qualitative data is **data without numerical values**.

It usually involves **descriptions, categories, labels, or classifications** rather than measurements or numbers.

It tells us *what something is like*, not *how much* of it there is.

Example(s)

- Interview answers describing a person's experience.
- Categories such as "happy," "neutral," or "sad."
- Types of user complaints listed as "navigation problem," "slow system," or "confusing layout."
- Photos or videos used to understand behaviour.

Mixed method (of data types)

What it is?

A mixed method uses **both qualitative data and quantitative data** in the same study.

This means researchers use qualitative data **as it is** (for deep understanding) and also **turn some of it into numbers** so it can be measured or compared.

It combines the strengths of both data types.

Example(s)

- Interview texts are read for meaning (qualitative), and then certain words or themes are counted (quantitative).
- A focus group discussion is analyzed for emotions and attitudes (qualitative), and then researchers code each emotion into categories and measure how often each appears (quantitative).
- Photos from observations are described (qualitative) and then classified into groups for statistical analysis (quantitative).

Interpretation

Main problem of interpretation

What it is?

The main problem of interpretation is that **many different mental-state explanations can fit the same behaviour**.

A single action can be explained by several **plausible beliefs and desires**, and these are called **alternative mental-state models**.

Because of this, interpretation is not always reliable.

How this problem can be faced

1. **Avoiding interpretation**

   Use methods that depend less on guessing mental states, such as studying behaviour, social patterns, or physiological signals.

2. **Improving interpretation**

   Gather more evidence, check for consistency, repeat questions, use multiple observers, and apply clear quality criteria to make interpretations more trustworthy.

Example(s)

- If someone rejects an offer, they might be angry, confused, proud, or following a social rule. Many explanations fit the same action.
- Observing more behaviour, asking questions, or checking with other observers helps decide which mental-state model is most likely.

Case study

What it is?

A case study is a type of research that focuses on **a small number of cases**, often even just one.

These cases are **chosen for a specific reason**, such as being unusual, important, or useful for answering a research question.

All cases usually show **the same outcome**, and the researcher studies them in depth.

Case studies **do not use experimental manipulation**; instead, they observe and analyze what has already happened.

The main goal is to **understand a specific case** as fully as possible.

Example(s)

- Studying one company to understand how it adapted to a crisis.
- Examining a single school known for unusually high performance.
- Researching one patient case to understand a rare medical condition.
- Analyzing a historical event to explain why it happened the way it did.

Multivariate study

What it is?

A multivariate study is research that uses a **large number of cases**, usually selected randomly or in another representative way so that they reflect the population.

The **dependent variable changes** across these cases, which allows researchers to compare different situations or conditions.

The study either includes **experimental manipulation** or tries to mimic it, for example through natural experiments.

The main goal is to **identify and understand general patterns** that apply to the broader population.

Example(s)

- Surveying thousands of people to study how income, education, and age affect voting behaviour.
- Comparing test scores across many schools to see how class size influences performance.
- Studying medical data from many patients to understand which factors increase disease risk.

Analytical narrative

What it is?

An analytical narrative is a type of case study that uses a **story-like explanation** to show how key actors, their preferences, and their decisions led to a particular outcome.

It focuses on identifying the **important actors**, their **main decision points**, and the logic behind their choices, often using ideas from game theory or rational choice.

The goal is to make the reasoning in the case clear and to generate explanations that can be evaluated.

Example(s)

- Telling the story of why a government chose a specific policy by describing the actors involved and the decisions they faced.
- Explaining why a company succeeded by outlining the sequence of strategic choices made by its leaders.
- Describing why a historical rebellion occurred by identifying the key groups and the decisions that shaped the event.

Qualitative comparative analysis

What it is?

Qualitative comparative analysis (QCA) is a method where several cases are compared to find out which conditions are **needed** for an outcome

to happen.

It helps researchers understand **when and why** a result occurs.

The findings can also be used later in larger, statistical studies.

Example(s)

- Comparing different successful strikes to see which conditions must always be present.
- Looking at several failing companies to find a factor that appears in every failure.
- Studying high-performing schools to see which condition is necessary for success.

Process tracing

What it is?

Process tracing is a type of case study that investigates **cause-and-effect inside one single case**, not many cases like Qualitative Comparative Analysis (QCA).

It tries to uncover the **exact sequence of steps**—the mechanism—that produced the outcome in that specific case.

Instead of looking for patterns across many cases or calculating average effects in a population, process tracing focuses on **how** and **why** the outcome happened in this one situation.

The goal is to identify the **causal mechanism**: the chain of events linking the cause to the final result.

Example(s)

- Studying a single political crisis to see which decisions, mistakes, or pressures directly caused the government to fall.

- Examining one company’s success by identifying each strategic move that led to its growth.
- Investigating a historical event, such as a revolution, by tracing the key moments and actions that built up to the final outcome.